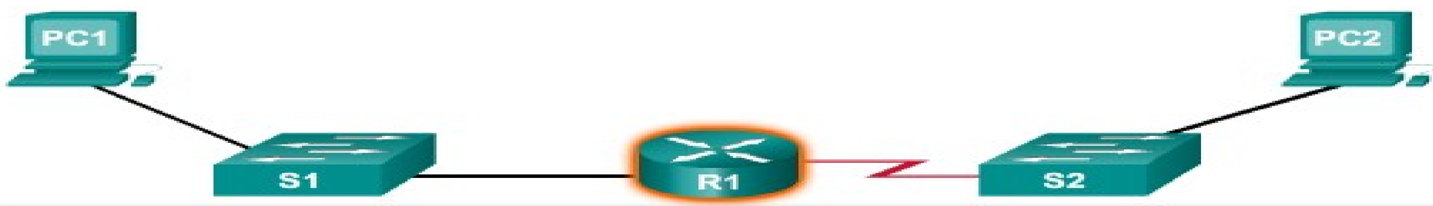
Chapter-2

Basic Router and switch Configuration

Why Routing?

- Routing is the method by which network devices **direct messages across networks** to arrive at the correct destination.
- The router is responsible for the **routing of traffic between networks.**

Routers Route Packets




```
R1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS
inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
Gateway of last resort is not set

C    192.168.1.0/24 is directly connected, FastEthernet0/0
C    192.168.2.0/24 is directly connected, Serial0/0
```

Cisco IOS command line interface (CLI) can be used to view the route table.

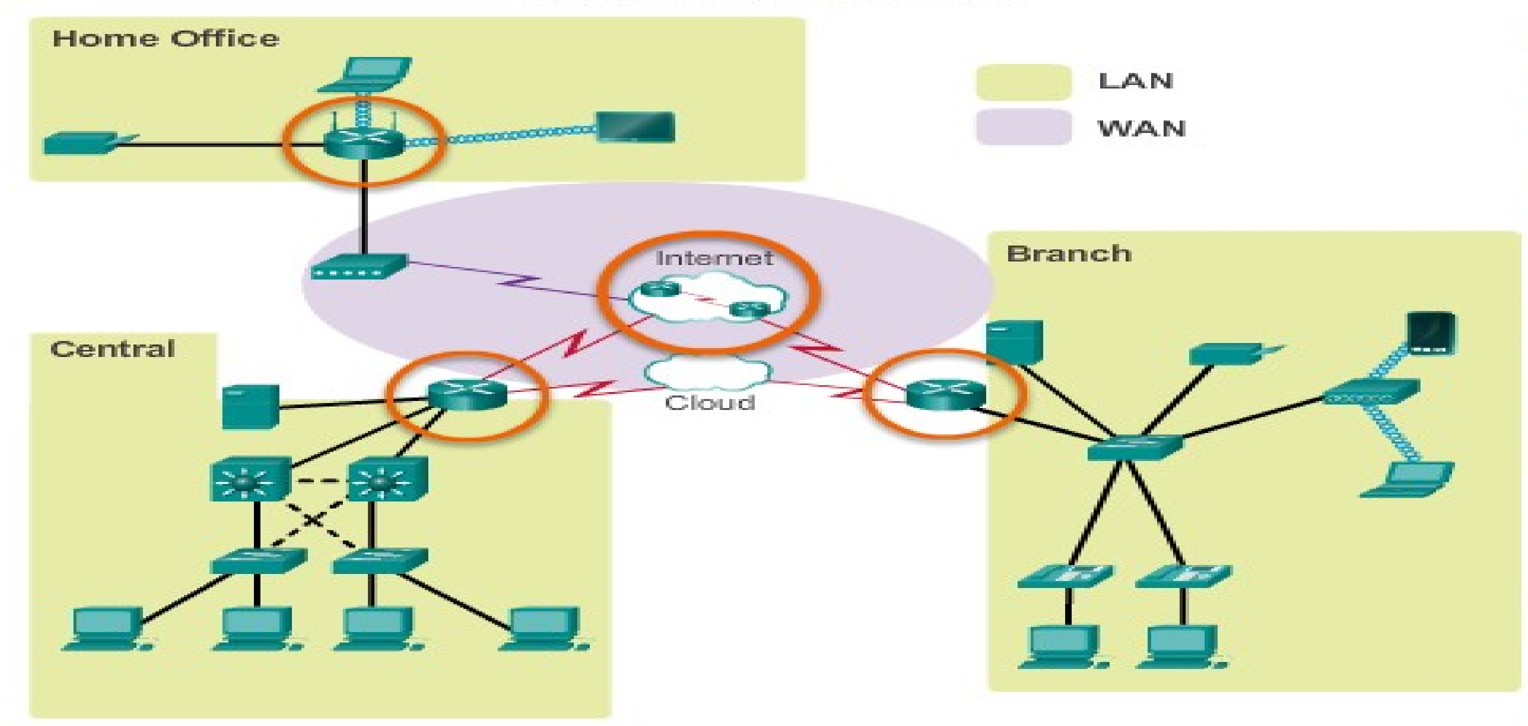
Routers are Computers

- Routers are specialized computers containing the following required components to operate:
  - ✓Central processing unit (CPU)
  - ✓Operating system (OS) - Routers use Cisco IOS
  - ✓Memory and storage (RAM, ROM, NVRAM, Flash, hard drive)
- Routers utilize the following memory:

Memory	Volatile / Non-Volatile	Stores
RAM	Volatile	• Running IOS • Running configuration file • IP routing and ARP tables • Packet buffer
ROM	Non-Volatile	• Bootup instructions • Basic diagnostic software • Limited IOS
NVRAM	Non-Volatile	• Startup configuration file
Flash	Non-Volatile	• IOS • Other system files

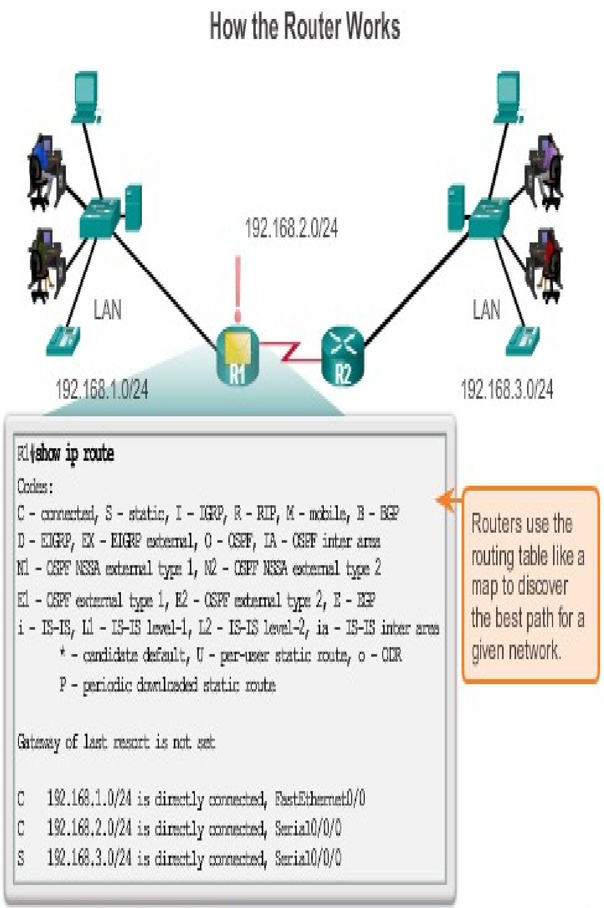
Routers Interconnect Networks

- Routers can connect multiple networks.
- Routers have **multiple interfaces**, each on a **different IP network**.



Routers Choose Best Paths

- Determine the best path to send packets
  - Uses its routing table to determine path
- Forward packets toward their destination
  - Forwards packet to interface indicated in routing table.
  - Encapsulates the packet and forwards out toward destination.



- Routers use static routes and dynamic routing protocols to learn about remote networks and build their routing tables.

Document Network Addressing

- ❑Network Documentation should include at least the following in a topology diagram and addressing table:

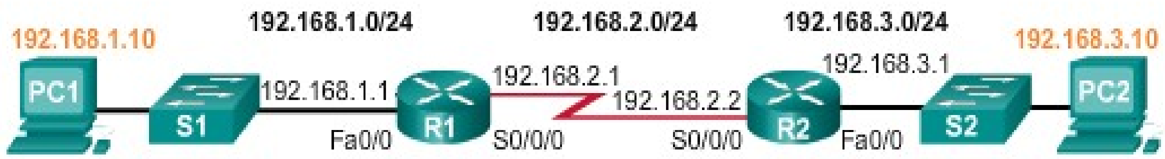
✓Device names

✓Interfaces

✓IP addresses and

✓ subnet mask

✓Default gateways



Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	Fa0/0	192.168.1.1	255.255.255.0	N/A
	S0/0/0	192.168.2.1	255.255.255.0	N/A
R2	Fa0/0	192.168.3.1	255.255.255.0	N/A
	S0/0/0	192.168.2.2	255.255.255.0	N/A
PC1	N/A	192.168.1.10	255.255.255.0	192.168.1.1
PC2	N/A	192.168.3.10	255.255.255.0	192.168.3.1

Best Path

- Best path is **selected by a routing protocol** based on the **metric value**.
- A **metric** is the value **used to measure** the distance to a given network.
- Best path to a network is the path with the **lowest metric**.
- **Routing Information Protocol (RIP)** - Hop count
- **Open Shortest Path First (OSPF)** - Cost based on cumulative bandwidth from source to destination
- **Enhanced Interior Gateway Routing Protocol (EIGRP)** - Bandwidth, delay, load, reliability

Load Balancing

- When a router has two or more paths to a destination with equal cost metrics, then the router forwards the packets using **both paths equally.**

Administrative Distance

- If multiple paths to a destination are configured on a router, the path installed in the routing table is the one with the best (lowest) Administrative Distance (AD).
- Administrative Distance is the "trustworthiness" of the route
- The Lower the AD the more trustworthy the route.

Route Source	Administrative Distance
Connected	0
Static	1
EIGRP summary route	5
External BGP	20
Internal EIGRP	90
IGRP	100
OSPF	110
IS-IS	115
External EIGRP	170
Internal BGP	200

The Routing Table

- Routing Table is a file stored in RAM that contains information about
  - Directly Connected Routes
  - Remote Routes
  - Network or Next hop Associations
- Show ip route command is used to display the contents of the routing table

Routing Table of R1


```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia -
       IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
Gateway of last resort is not set
```

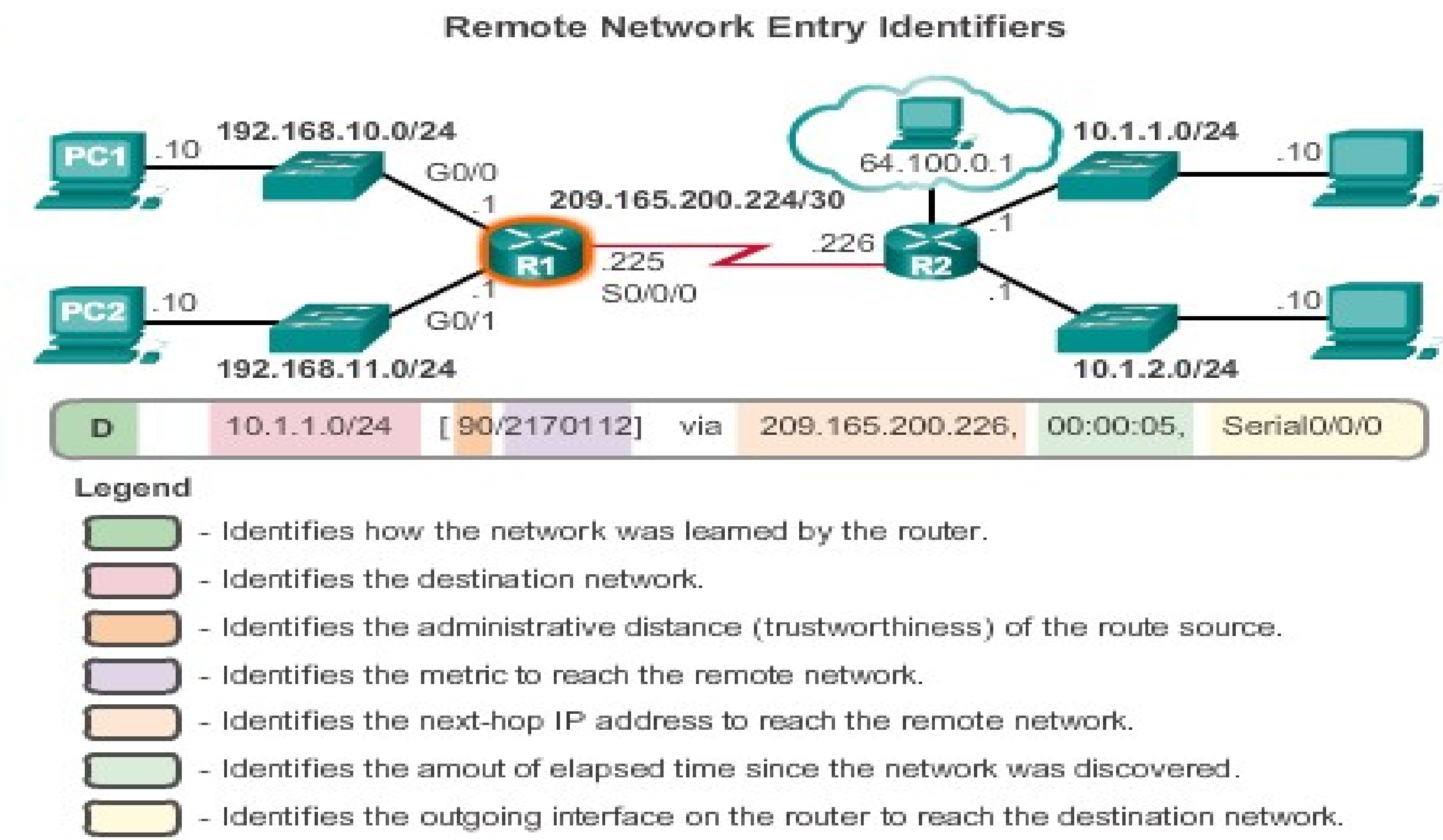
The routing table

- A route has four main components:
  I. Destination value
  II. Mask
  III. Gateway or interface address
  IV. Route cost or metric

- To direct a message to the correct destination, the router looks at the destination IP address in the packet and then looks for a matching route in the routing table.

The Routing Table

- Interpreting the entries in the routing table.



- There are several types of routes that can appear in the routing table:

❑**Directly-Connected Routes**

- The router stores directly attached local network addresses as connected routes in the routing table.
- For Cisco routers, these routes are identified in the routing table with the **prefix C**.
- These routes are automatically updated whenever the interface is reconfigured or shutdown.

❑Static Routes

- A network administrator can manually configure a static route to a specific network.
- A static route does not change until the administrator manually reconfigures it.
- These routes are identified in the routing table with the prefix S.

❑Dynamically-Updated Routes (Dynamic Routes)

- Dynamic routes are automatically created and maintained by routing protocols.
- Dynamically-updated routes are identified in the routing table with the prefix the type of routing protocol that created the route,

For example **R** is used for the Routing Information Protocol (RIP).

❑Default Route

- The default route is a type of static route which specifies a gateway to use when the routing table does not contain a path to use to reach the destination network.

- If the router does not have a route to a specific network in its routing table, a default route can be configured to tell the router how to forward the packet.

- The default route is used by the router only if the router does not know where to send a packet.

Basic configuration

Set Device Name

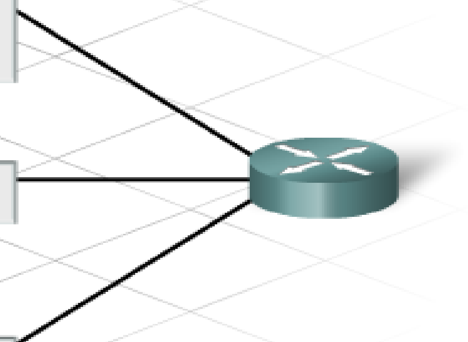
```
Router(config)#hostname TokyoRouter
TokyoRouter(config)#
```

Enable Password

```
Router(config)#enable password san-fran
```

Enable Encrypted Password

```
Router(config)#enable secret password123
```

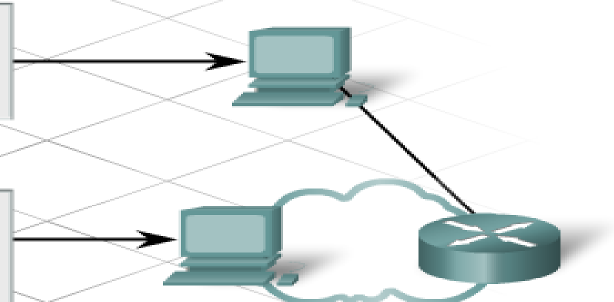


Console Password

```
Router(config)#line console 0
Router(config-line)#password cisco
Router(config-line)#login
```

Virtual Terminal Password

```
Router(config)#line vty 0 4
Router(config-line)#password cisco
Router(config-line)#login
```



Perform Password Encryption

```
Router(config)#service password-encryption
```

Basic configuration

❑Configuring Router interface

- There are many different types of interfaces available in router.
- Serial and Ethernet interfaces are the most common.
- Local network connections use Ethernet interfaces.
- WAN connections require the use of a serial connection through a ISP.
- Unlike Ethernet interfaces, serial interfaces require a clock signal to control the timing of the communications, this is known as a clock rate.
- The available clock rates in bits per second are 1200, 2400, 9600, 19200, 38400, 56000, 64000, 72000, 125000, 148000, 500000, 800000, 1000000, 1300000, 2000000, or 4000000.

- Data Communications Equipment (DCE) devices provides the clock rate.
- Router can be configured as DCE devices, if necessary.
- The steps to configure an interface include:

1. Specify the type of interface and the interface port number
2. Specify a description of the interface……optional
3. Configure the interface IP address and subnet mask
4. Set the clock rate, if configuring a serial interface as a DCE
5. Enable the interface

```
Router (config)#interface serial 0/0/0
Router (config-if)#description connection to Router2
Router (config-if)#ip address 192.168.1.125 255.255.255.0
Router (config-if)#clock rate 64000
Router (config-if)#no shutdown
```

Configure fast Ethernet interface

Router(config)#int fa0/0

Router(config-if)#ip address 192.168.10.1 255.255.255.0

Router(config-if)#no shut down

Static Rout

- A router can learn about remote networks in one of two ways:
  - **Manually** - Remote networks are manually entered into the route table using static routes.
  - **Dynamically** - Remote routes are automatically learned using a dynamic routing protocol.

❑ Why Use Static Routing?

- Static routing provides some advantages over dynamic routing, including:

✓ Better security, because of not advertised over the network.

✓ Use less bandwidth than dynamic routing protocols, no CPU cycles are used to calculate and communicate routes.

✓ The path is known to send data.

Static route

- Static routing has the following **disadvantages**:

✓Initial configuration and maintenance is time-consuming.

✓Configuration is error-prone, especially in large networks.

✓Administrator intervention is required to maintain changing route information.

✓Does not scale well with growing networks; maintenance becomes bulky.

✓Requires complete knowledge of the whole network for proper implementation.

When to Use Static Routes

We can use Static routing in the following condition:

- In smaller networks that are not expected to grow significantly.

- Routing to and from stub networks:

✓ A stub network is a network accessed by a single route, and the router has no other neighbors.

- In default route.

Configuring Static Routes

- The steps to configure a static route on a router are as follows:

1. Connect to the router using a console cable.

2. Open a HyperTerminal window to connect with the first router that you want to configure.

3. Enter privileged mode, by typing enable at the Router1> prompt.

    Router1>enable

    Router1#

4. Enter global configuration mode.

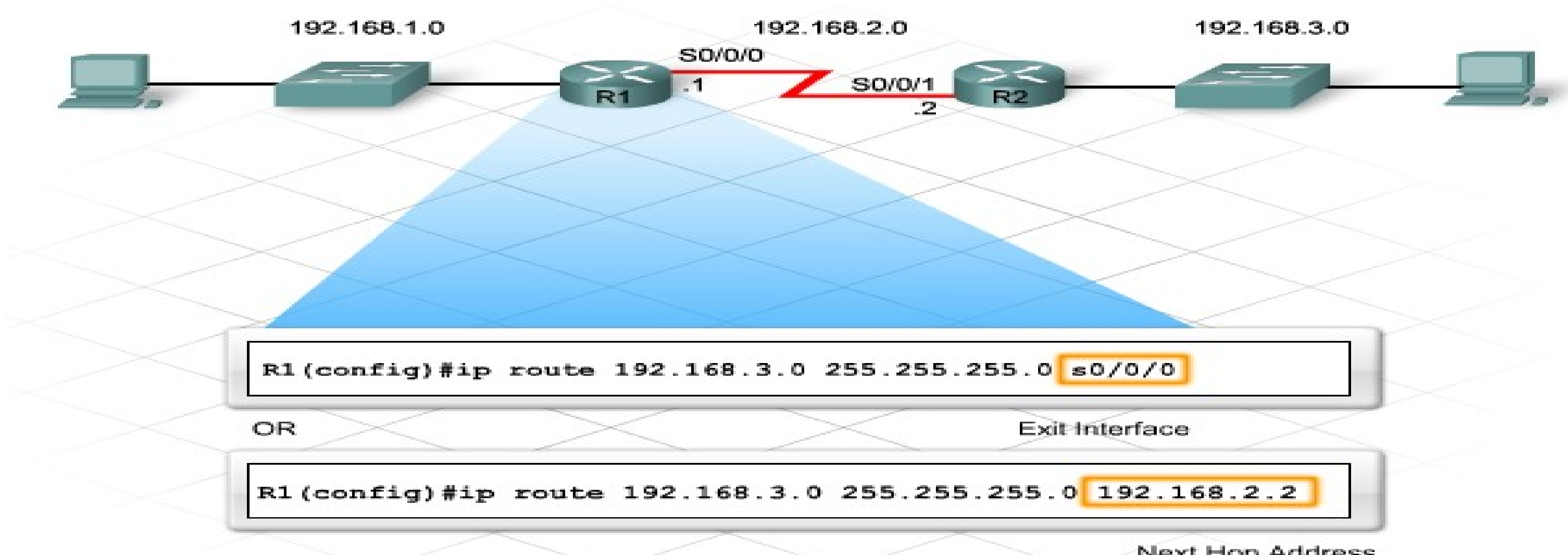
    Router1#config terminal

    Router1(config)#

Configuring Static Routes

5. Use the ip route command to configure the static route, with the following format:

ip route [destination_network] [subnet_mask] [next hop IP/exit interface]



Static route

- Static routes configured with **exit interface** required two steps, we can call it a recursive lookup.
- First it check destination Ip address, if the destination IP available in routing table.
- Second it find interface, in which interface it connect, if the exit interface disabled, static route disappear from routing table.

Static route

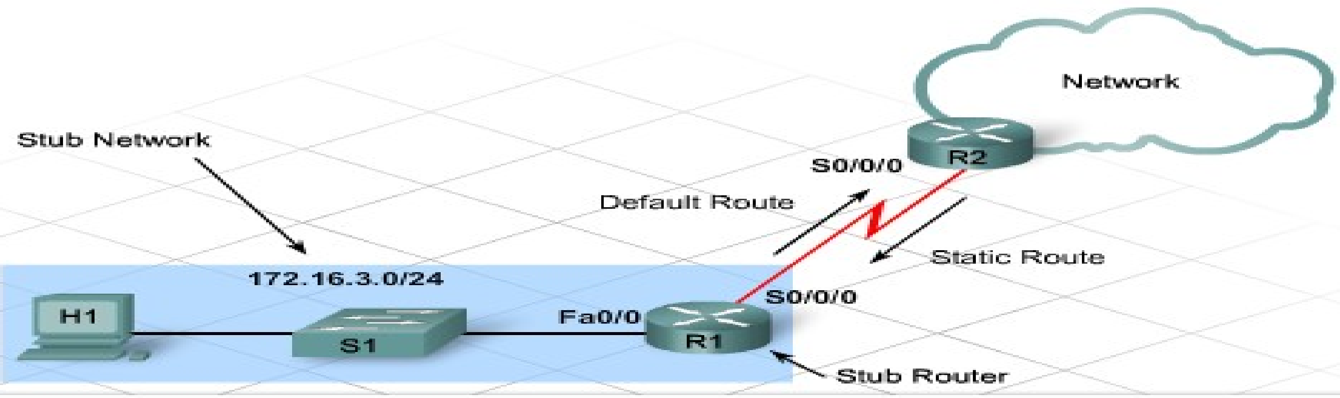
- To enable two-way communication with a host on network 192.168.16.0, the administrator also configures a static route on Router 2.
- Since static routes are configured manually, network administrators must **add** and **delete** static routes to reflect any changes in network topology.
- In a large network, the manual maintenance of routing tables could require significant administrative time.
- For this reason, **larger networks** generally use dynamic routing rather than static routes.

Default Routes

- Routing tables cannot contain routes to every possible Internet site.
- As routing tables grow in size, they require more RAM and processing power.
- A special type of static route, called a default route, specifies a gateway to use when the routing table does not contain a path to a destination.
- The command to create a default route is similar to the command used to create static route.
- The network address and subnet mask are both specified as 0.0.0.0, making it a quad zero route.
- The command uses either the next-hop address or the exit interface parameters.

Default Route

- As long as a better match does not exist, the router uses the default static route.
- This information appears in the routing tables of all routers.




```
R1(config)#ip route 0.0.0.0 0.0.0.0 s0/0/0
R1(config)#end

R1#show ip route
<output ommitted>

        Gateway of last resort is 0.0.0.0 to network 0.0.0.0

        172.16.0.0/24 is subnetted, 2 subnets
C       172.16.2.0 is directly connected, Serial0/0/0
C       172.16.3.0 is directly connected, FastEthernet0/0
S       0.0.0.0/0 is directly connected, Serial0/0/0
```

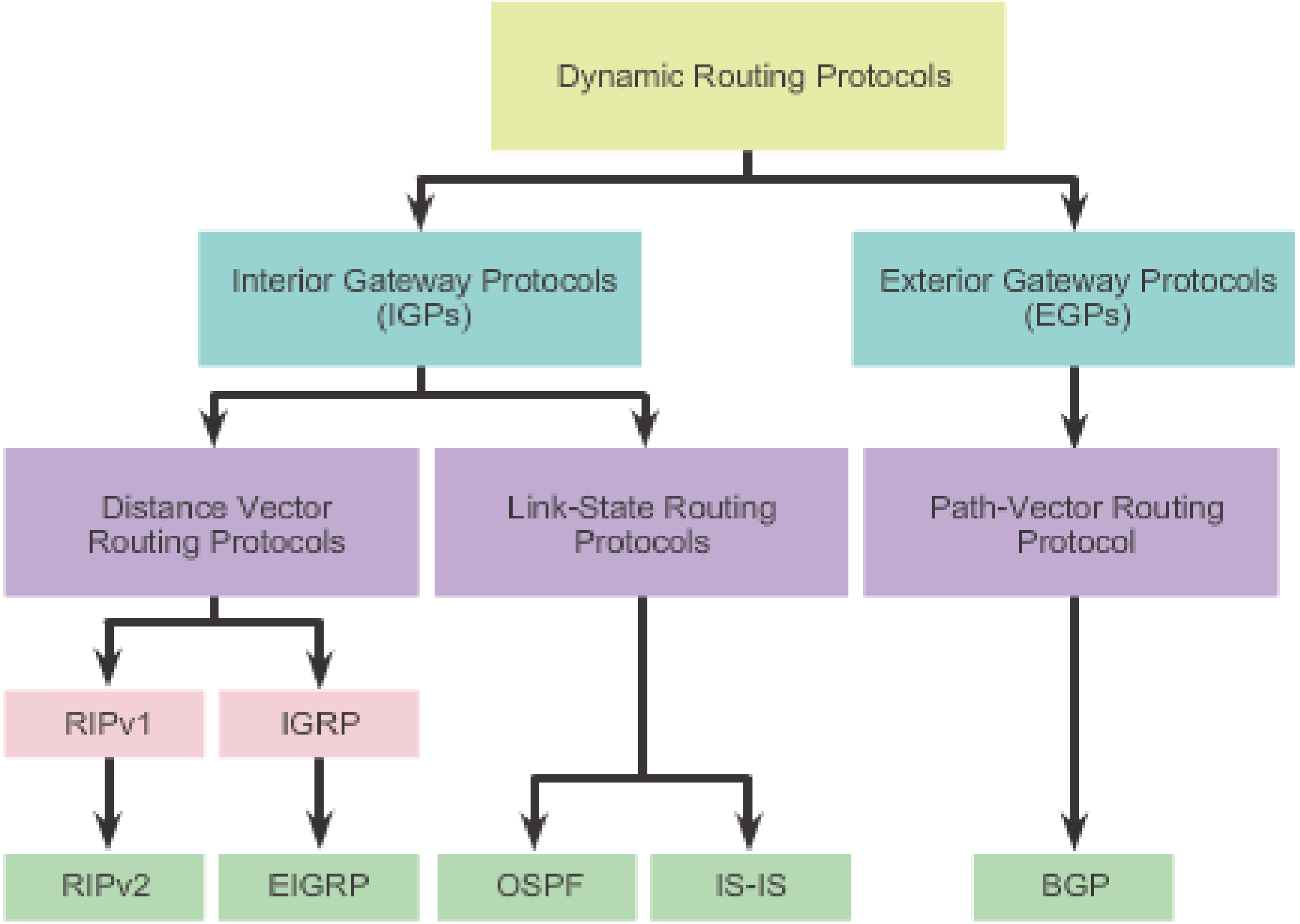
Dynamic Route

- Problems with cables and hardware failures can make destinations unreachable.
- Routers need a way to quickly update routes that does not depend on the administrator to make changes.
- Routers use routing protocols to dynamically manage information received from their own interfaces and from other routers.
- Dynamic routing makes it possible to avoid the time-consuming and exacting process of configuring static routes.
- Dynamic routing enables routers to react to changes in the network and to adjust their routing tables accordingly, without the intervention of the network administrator.

Dynamic Route

- A dynamic routing protocol learns all available routes, places the best routes into the routing table, and removes routes when they are no longer valid.
- The method that a routing protocol uses to determine the best route to a destination network is called a routing algorithm.
- There are two main classes of routing algorithms:
    - Distance vector
    - Link state
- Each type uses a different method for determining the best route to a destination network.
- When all the routers in a network have updated their tables to reflect the new route, the routers are said to have **converged**.

Routing Protocols Classification



Distance vector routing protocol

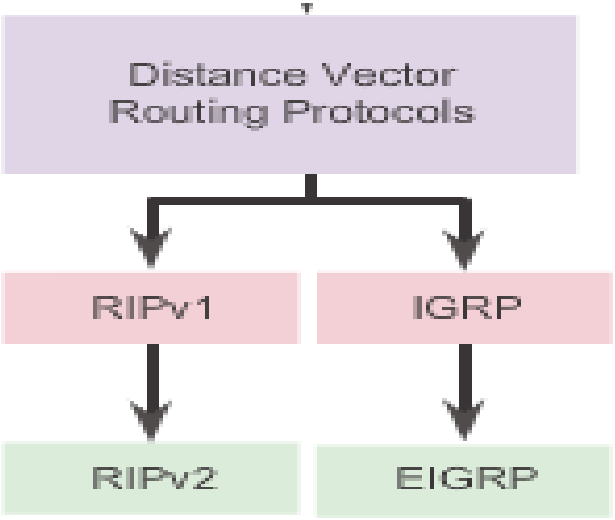
- The distance vector routing algorithm passes periodic copies of a routing table from router to router.
- The distance vector algorithm evaluates the route information it receives from other routers in terms of two basic criteria:
  - **Distance** - How far away is the network from this router?
  - **Vector** - In what direction should the packet be sent to reach this network?
- The distance component of a route is expressed in terms of cost, or metric, that can represent:

Number of hops	Administrative cost
Bandwidth	Transmission speed
Likelihood of delays	Reliability

Distance vector routing protocol

- The vector or direction component of a route is the address of the next hop along the path to the network named in the route.
- Each router that uses distance vector routing communicates its routing information to its neighbors.
- Neighbor routers share a directly connected network. The interface that leads to each directly connected network has a distance of 0.
- A router running a distance vector protocol does not know the entire path to a destination; it only knows the distance to the remote network and the direction, or vector.

- Protocol under Distance vector protocol are, RIPv1&v2, IGRP and EIGRP.



Distance vector routing protocols

- Share updates between neighbors
- Not aware of the network topology
- Some send periodic updates to broadcast IP 255.255.255.255 even if topology has not changed
- Updates consume bandwidth and network device CPU resources
- RIPv2 and EIGRP use multicast addresses
- EIGRP will only send an update when topology has changed

Distance Vector Algorithm

Purpose of Routing Algorithms

- Sending and receiving updates
- Calculate best path and install route
- Detect and react to topology changes

- RIP uses the Bellman-Ford algorithm as its routing algorithm
- IGRP and EIGRP use the Diffusing Update Algorithm (DUAL) routing algorithm developed by Cisco

Link-state routing protocols

- Enterprise networks and ISPs use link-state protocols because of their hierarchical design and ability to scale for large networks.
- Distance vector routing protocols are usually not the right choice for a complex enterprise network.
- Link-state routing protocols, such as OSPF, do not send frequent periodic updates of the entire routing table.
- Instead, after the network converges, a link-state protocol sends an update only when a change in the topology occurs, such as a link going down.

- **Compared with distance** vector protocols, link-state routing protocols:

✓Requires more complex network planning and configuration

✓Requires increased router resources

✓Requires more memory for storing multiple tables

✓Requires more CPU and processing power for the complex routing calculations

Purpose of dynamic routing protocols

- Purpose of dynamic routing protocols includes:
  - Discovery of remote networks
  - Maintaining up-to-date routing information
  - Choosing the best path to destination networks
  - Ability to find a new best path if the current path is no longer available.
- Main components of dynamic routing protocols include:
- **Data structures** - Routing protocols typically use tables or databases for its operations. This information is kept in RAM.
- **Routing protocol messages** - Routing protocols use various types of messages to discover neighboring routers, exchange routing information.
- **Algorithm** - Routing protocols use algorithms for facilitating routing information for **best path determination**.

- Advantages of dynamic routing
  - Automatically share information about remote networks
  - Determine the best path to each network and add this information to their routing tables
  - Compared to static routing, dynamic routing protocols require less administrative overhead.
  - Help the network administrator manage the time-consuming process of configuring and maintaining static routes

- Disadvantages of dynamic routing
  - Dedicate part of a routers resources for protocol operation, including CPU time and network link bandwidth

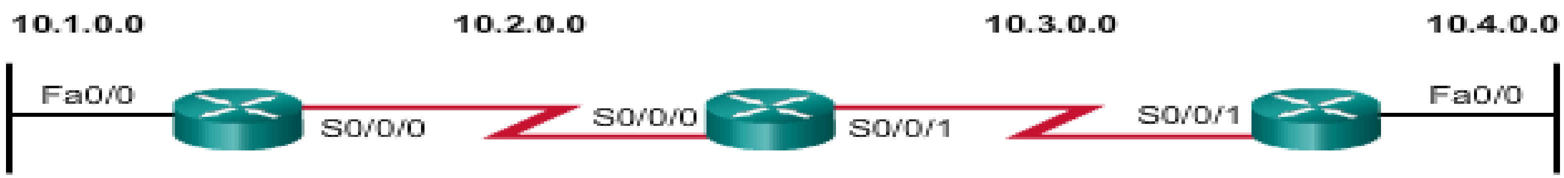
Dynamic Routing Protocol Operation

- In general, the operations of a dynamic routing protocol can be described as follows:

1. The router sends and receives routing messages on its interfaces.
2. The router shares routing messages and routing information with other routers that are using the same routing protocol.
3. Routers exchange routing information to learn about remote networks.
4. When a router detects a topology change the routing protocol can advertise this change to other routers.

Start

Directly Connected Networks Detected



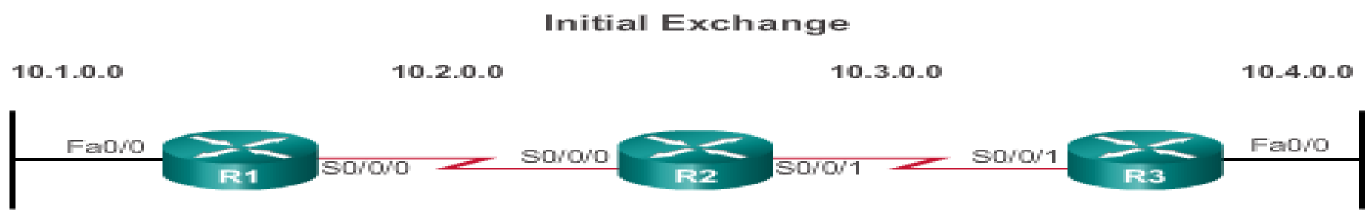
Network	Interface	Hop
10.1.0.0	Fa0/0	0
10.2.0.0	S0/0/0	0

Network	Interface	Hop
10.2.0.0	S0/0/0	0
10.3.0.0	S0/0/1	0

Network	Interface	Hop
10.3.0.0	S0/0/1	0
10.4.0.0	Fa0/0	0

- R1 adds the 10.1.0.0 network available through interface Fast Ethernet 0/0 and 10.2.0.0 is available through interface Serial 0/0/0.
- R2 adds the 10.2.0.0 network available through interface Serial 0/0/0 and 10.3.0.0 is available through interface Serial 0/0/1.
- R3 adds the 10.3.0.0 network available through interface Serial 0/0/1 and 10.4.0.0 is available through interface FastEthernet 0/0.

Network Discovery



Network	Interface	Hop
10.1.0.0	Fa0/0	0
10.2.0.0	S0/0/0	0
10.3.0.0	S0/0/0	1

Network	Interface	Hop
10.2.0.0	S0/0/0	0
10.3.0.0	S0/0/1	0
10.1.0.0	S0/0/0	1
10.4.0.0	S0/0/1	1

Network	Interface	Hop
10.3.0.0	S0/0/0	0
10.4.0.0	Fa0/0	0
10.2.0.0	S0/0/1	1

R1:

- Sends an update about network 10.1.0.0 out the Serial0/0/0 interface
- Sends an update about network 10.2.0.0 out the FastEthernet0/0 interface
- Receives update from R2 about network 10.3.0.0 with a metric of 1
- Stores network 10.3.0.0 in the routing table with a metric of 1

Exchanging the Routing Information

Network	Interface	Hop
10.1.0.0	Fa0/0	0
10.2.0.0	S0/0/0	0
10.3.0.0	S0/0/0	1
10.4.0.0	S0/0/0	2

Network	Interface	Hop
10.2.0.0	S0/0/0	0
10.3.0.0	S0/0/1	0
10.1.0.0	S0/0/0	1
10.4.0.0	S0/0/1	1

Network	Interface	Hop
10.3.0.0	S0/0/1	0
10.4.0.0	Fa0/0	0
10.2.0.0	S0/0/1	1
10.1.0.0	S0/0/1	2

R1:

- Sends an update about network 10. 1. 0. 0 out the Serial 0/0/0 interface
- Sends an update about networks 10. 2. 0. 0 and 10. 3. 0. 0 out the FastEthernet0/0 interface
- Receives an update from R2 about network 10. 4. 0. 0 with a metric of 2
- Stores network 10. 4. 0. 0 in the routing table with a metric of 2
- Same update from R2 contains information about network 10. 3. 0. 0 with a metric of 1.

RIP(Routing Information Protocol)

- Routing Information Protocol (RIP) was the first distance vector routing protocol and supported by most router.
- By default RIP broadcasts its routing updates out all active interfaces every 30 seconds.
- RIPv1 is a classful routing protocol. It **automatically summarizes** subnets to the classful boundary and does not send subnet mask information in the update.
  - Hop-count metric
  - 15-hop maximum
  - Default 30-second update interval
  - Updates using UDP port 520
  - Administrative distance of 120

- RIP- sends a request message to neighbor device to get routing table of neighbor device.

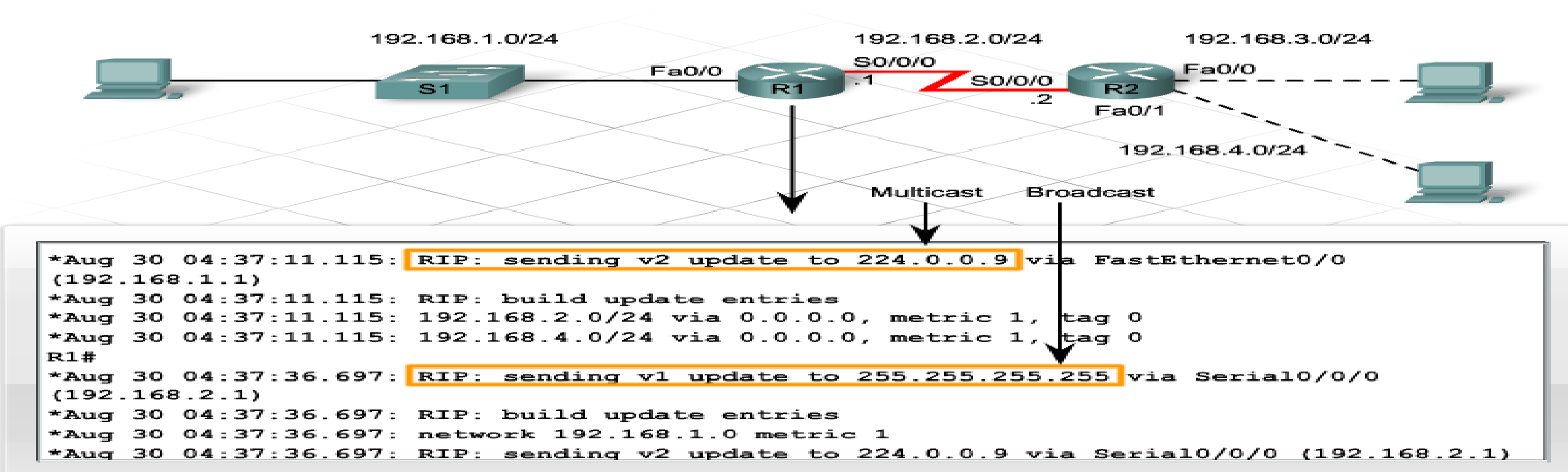
- The receiving router evaluates each route entry based on the following criteria:

✓ If a route entry is new, the receiving router add route in the routing table.

✓ If the route is already in the table and the entry comes from a different source, the routing table replaces the existing entry if the **new entry has a better hop count**.

✓ If the route is already in the table and the entry comes from the **same source**, it replaces the existing entry even if the metric is not better.

- RIPv2 is a **classless** routing protocol.
- RIPv2 also has the ability to turn off automatic summarization of routes.
- RIP v1 **broadcasts** these updates to **255.255.255.255** and RIP V2 **multicast** its update to **224.0.0.9.**
- Multicasts take up less network **bandwidth** than broadcasts.
- Devices that are not configured for RIPv2 discard multicasts at the Data Link Layer.



RIPv1 and RIPv2 Routing Updates

Rip configuration

- Before configuring RIP, **assign IP addresses** and **masks** to all interfaces that participate in routing.
- Set the **clock rate** where necessary on **serial links.**
- The basic RIP configuration consists of:-

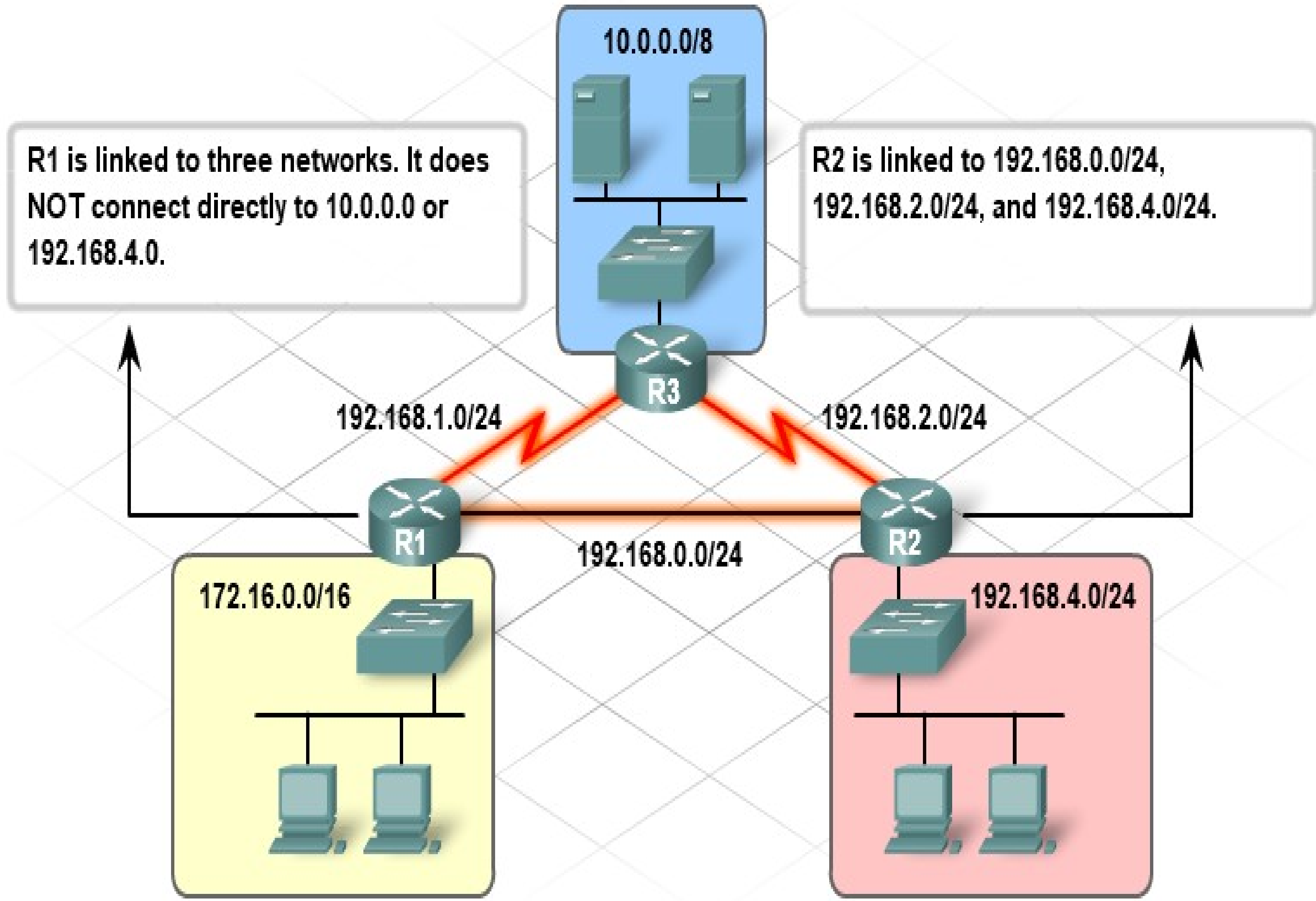
RIPv1

- **Router(config)#router rip**
- **Router(config-router)#network [network address]**

RIPv2

- **Router(config)#router rip**
- **Router(config)#version 2**
- **Router(config-router)#network [network address]**

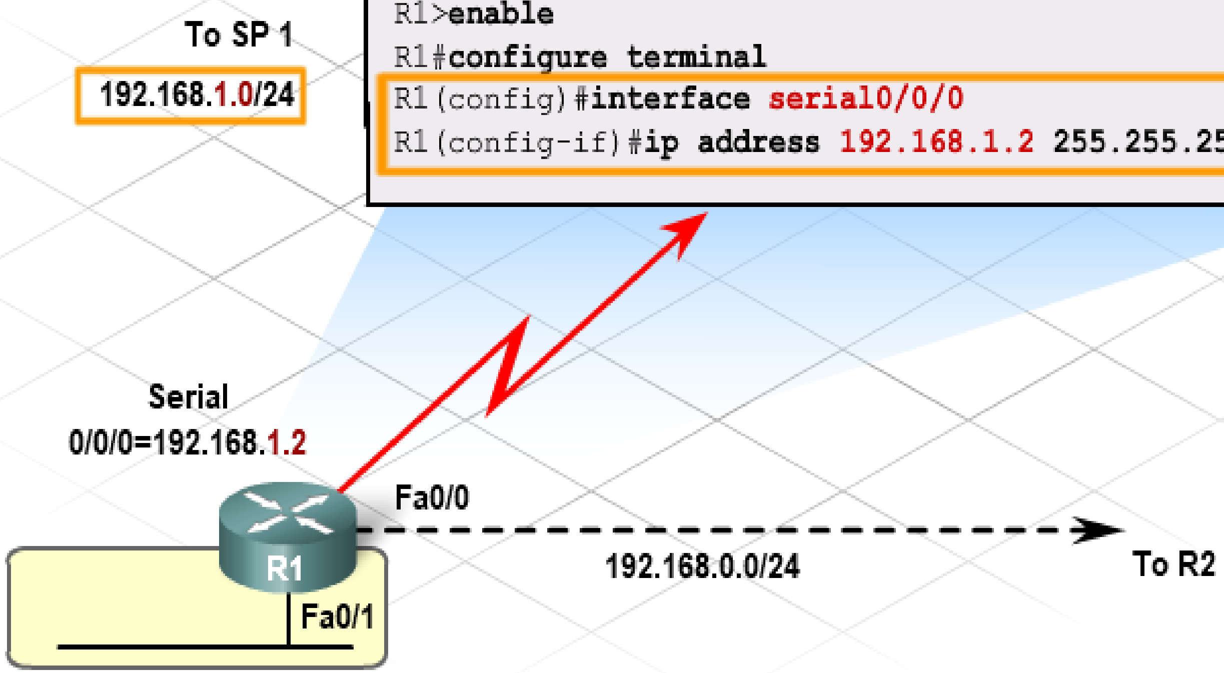
Determining the Network Configuration



Configure the Serial Interface Address

The R1 router has three interfaces to configure. Serial 0/0/0 links to the R3 router, Fastethernet 0/0 links to R2, and Fastethernet 0/1 links to the 172.16.0.0/16 production network. Configure Serial 0/0/0 first.

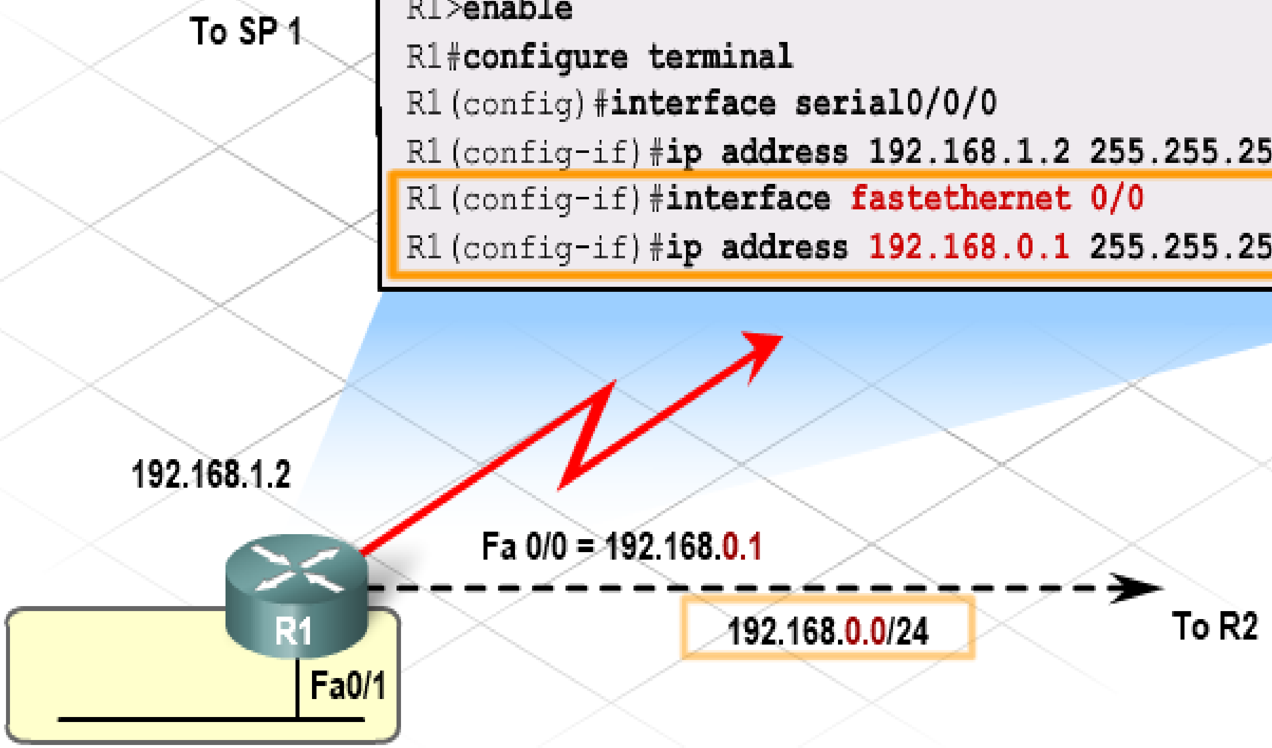
```
R1>enable
R1#configure terminal
R1(config)#interface serial0/0/0
R1(config-if)#ip address 192.168.1.2 255.255.255.0
```



Configure the Fast Ethernet Interface

For each of the three interfaces, assign a previously unused IP address from the network that the interface connects to. Fastethernet 0/0 points to R2 and is on the 192.168.0.0/24 network. Assign this interface the first useable IP address from that network.

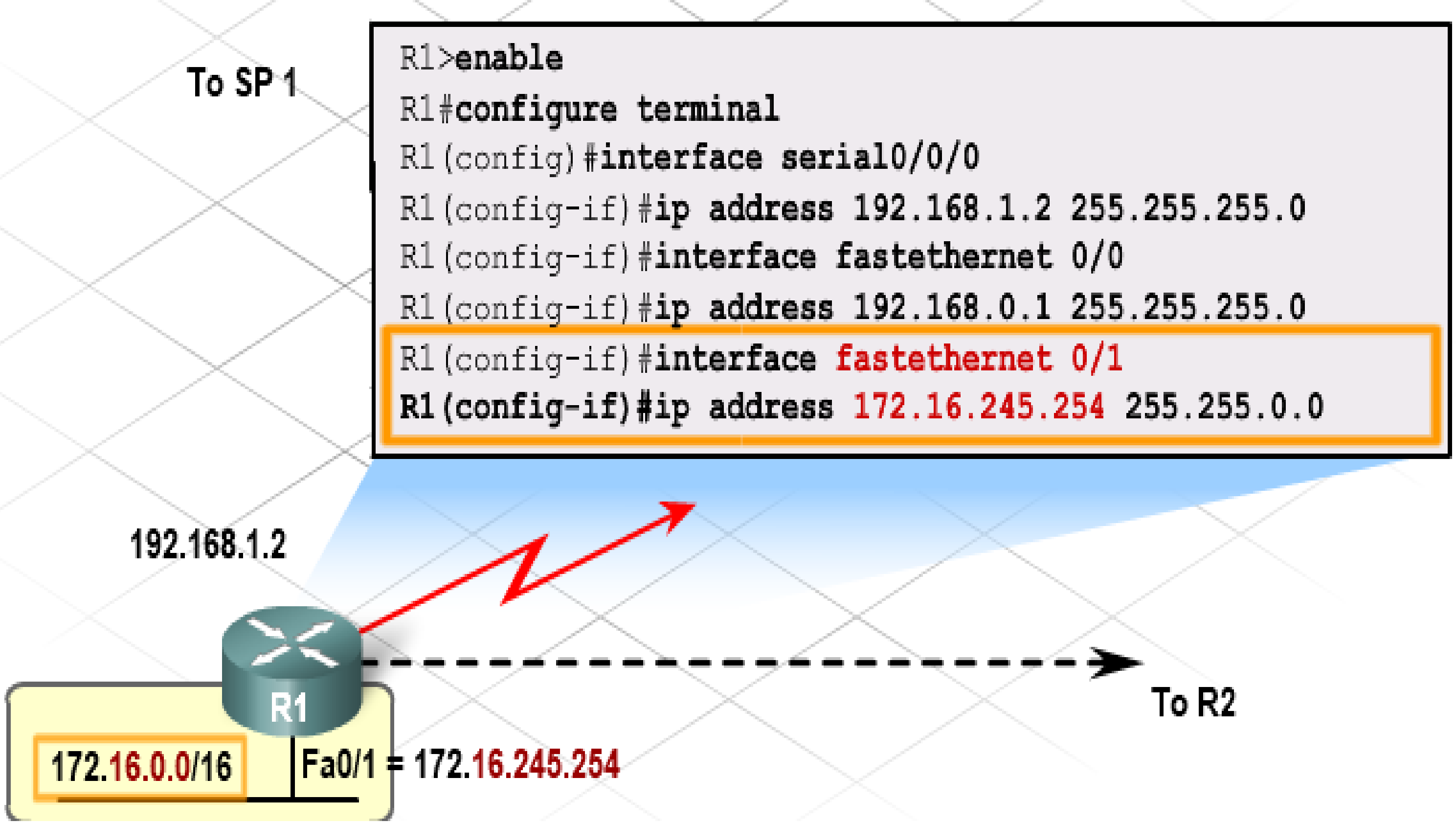
```
R1>enable
R1#configure terminal
R1(config)#interface serial0/0/0
R1(config-if)#ip address 192.168.1.2 255.255.255.0
R1(config-if)#interface fastethernet 0/0
R1(config-if)#ip address 192.168.0.1 255.255.255.0
```



Configure and Verify the IP Addresses

Configure the last interface on R1.

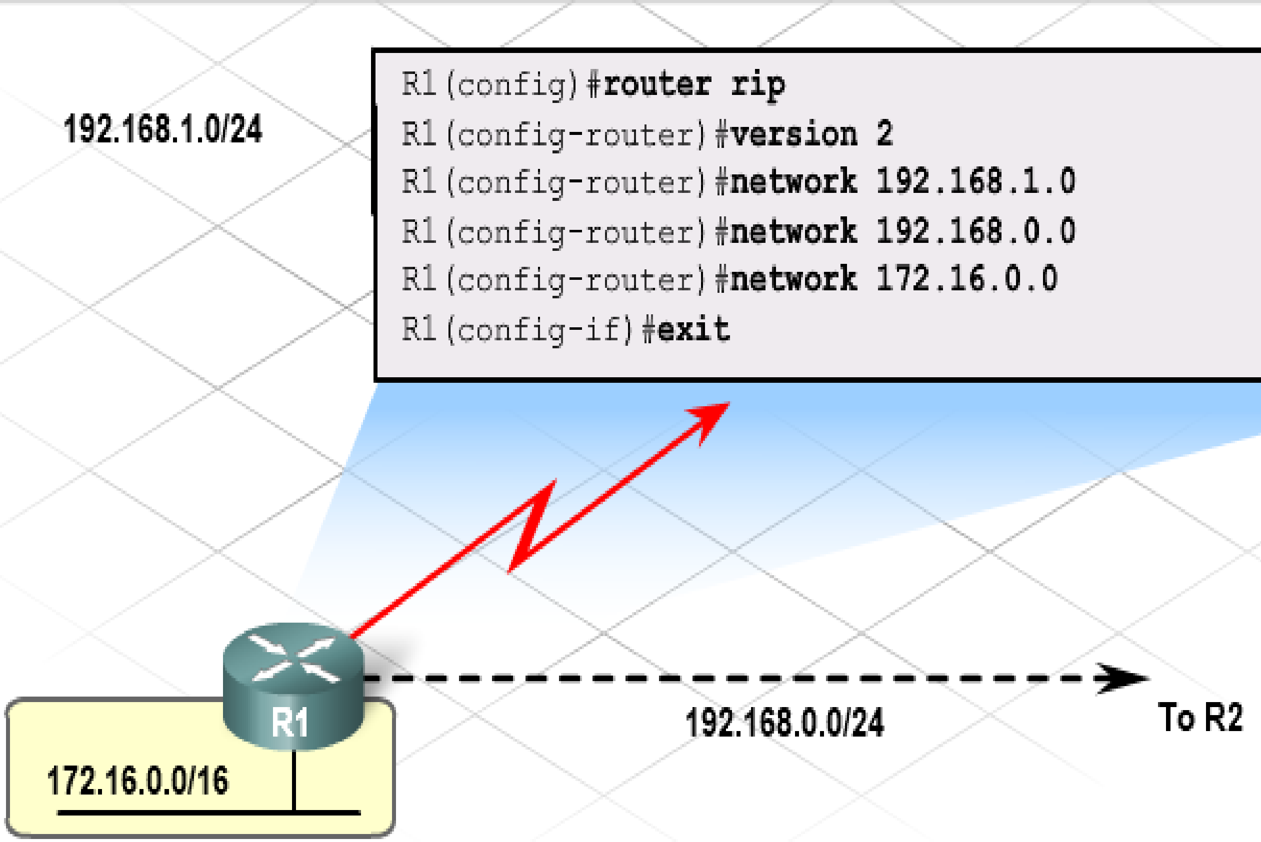
```
R1>enable
R1#configure terminal
R1(config)#interface serial0/0/0
R1(config-if)#ip address 192.168.1.2 255.255.255.0
R1(config-if)#interface fastethernet 0/0
R1(config-if)#ip address 192.168.0.1 255.255.255.0
R1(config-if)#interface fastethernet 0/1
R1(config-if)#ip address 172.16.245.254 255.255.0.0
```



Configure and Verify the RIP

Specify RIP version 2 and tell the router which networks it can advertise. Use the network command for each directly connected network. R1 connects to three networks, so those networks are entered here.

```
R1(config)#router rip
R1(config-router)#version 2
R1(config-router)#network 192.168.1.0
R1(config-router)#network 192.168.0.0
R1(config-router)#network 172.16.0.0
R1(config-if)#exit
```



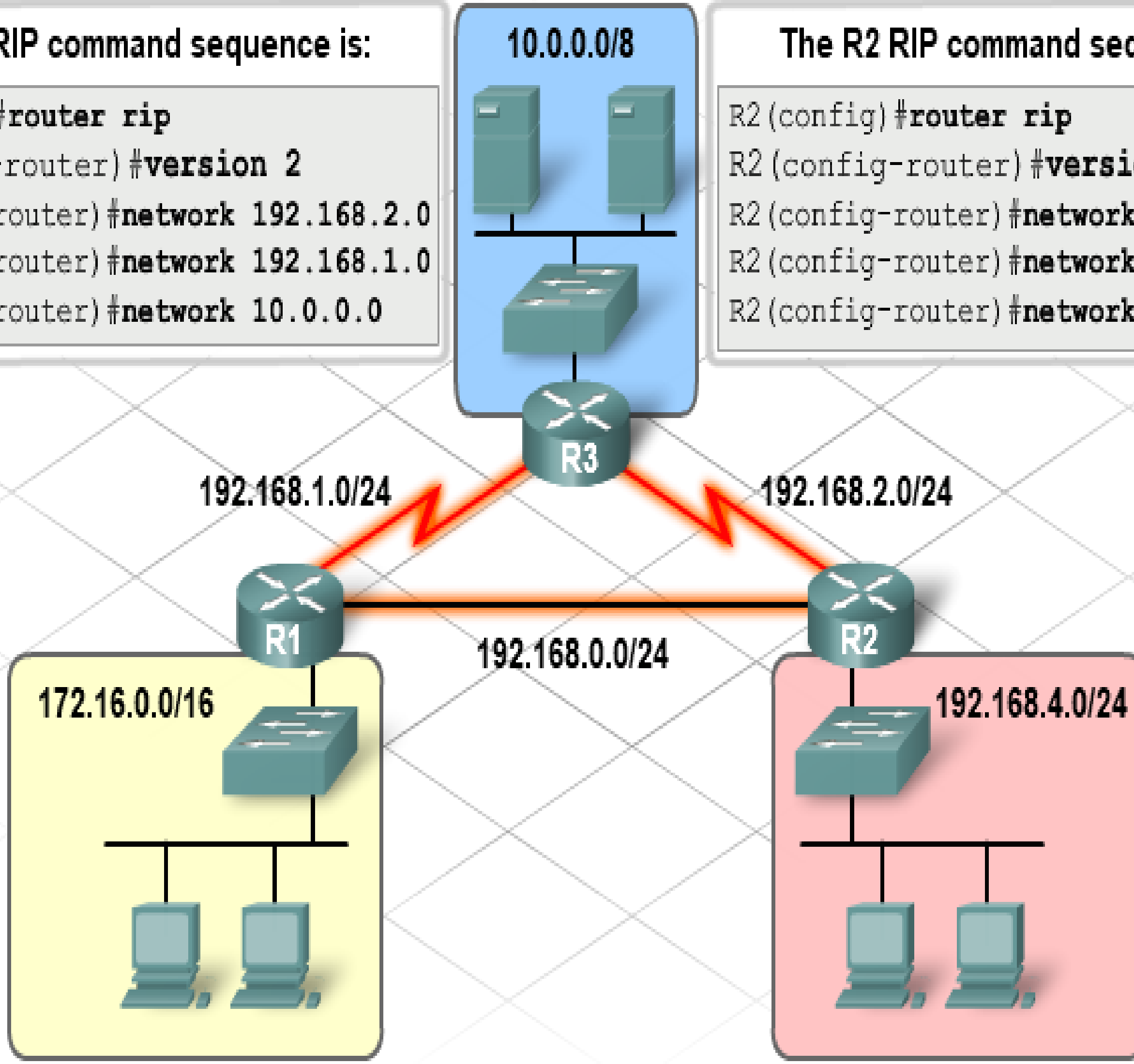
Complete the Configuration of All Routers

The R3 RIP command sequence is:

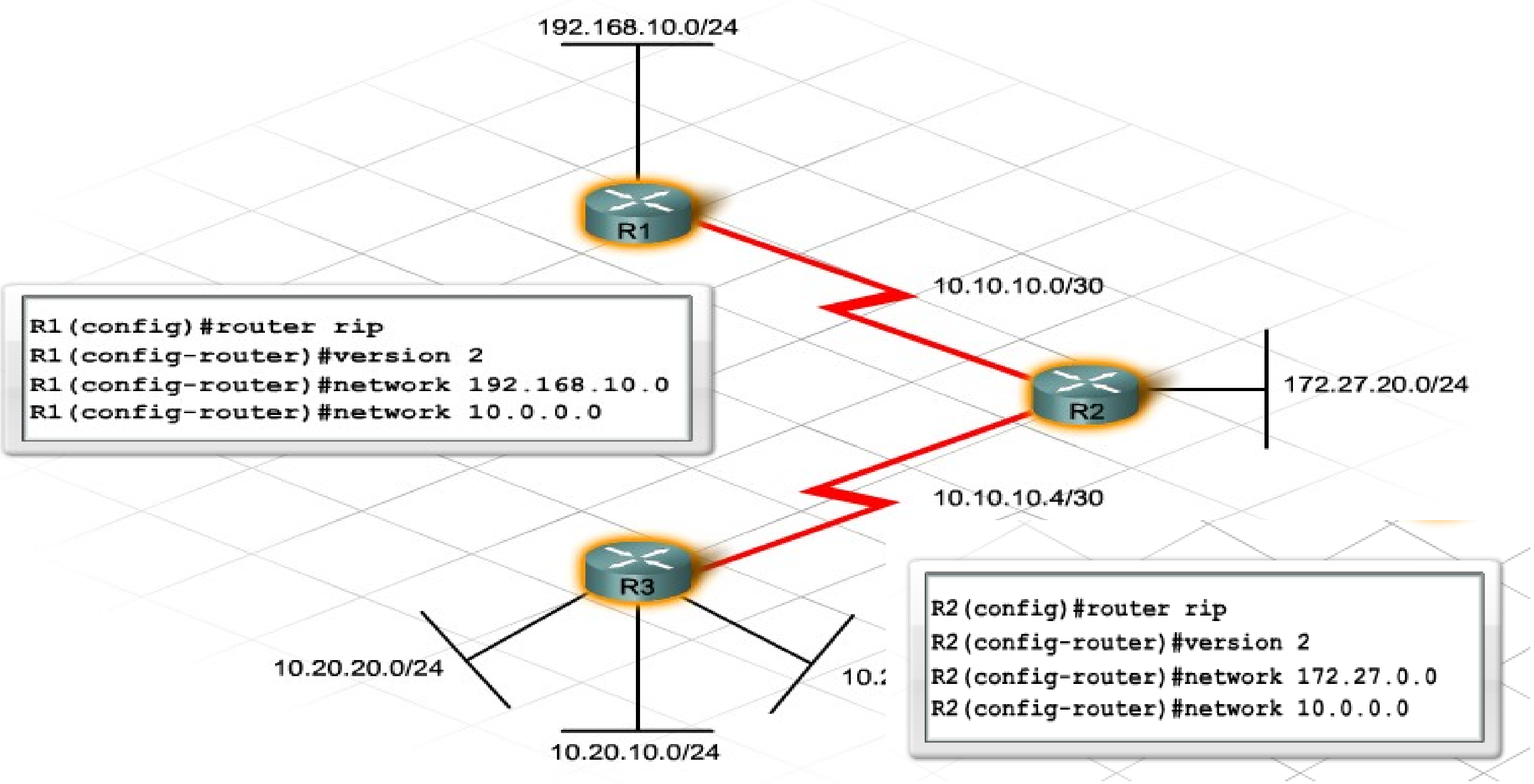
```
R3(config)#router rip
R3(config-router)#version 2
R3(config-router)#network 192.168.2.0
R3(config-router)#network 192.168.1.0
R3(config-router)#network 10.0.0.0
```

The R2 RIP command sequence is:

```
R2(config)#router rip
R2(config-router)#version 2
R2(config-router)#network 192.168.2.0
R2(config-router)#network 192.168.0.0
R2(config-router)#network 192.168.4.0
```



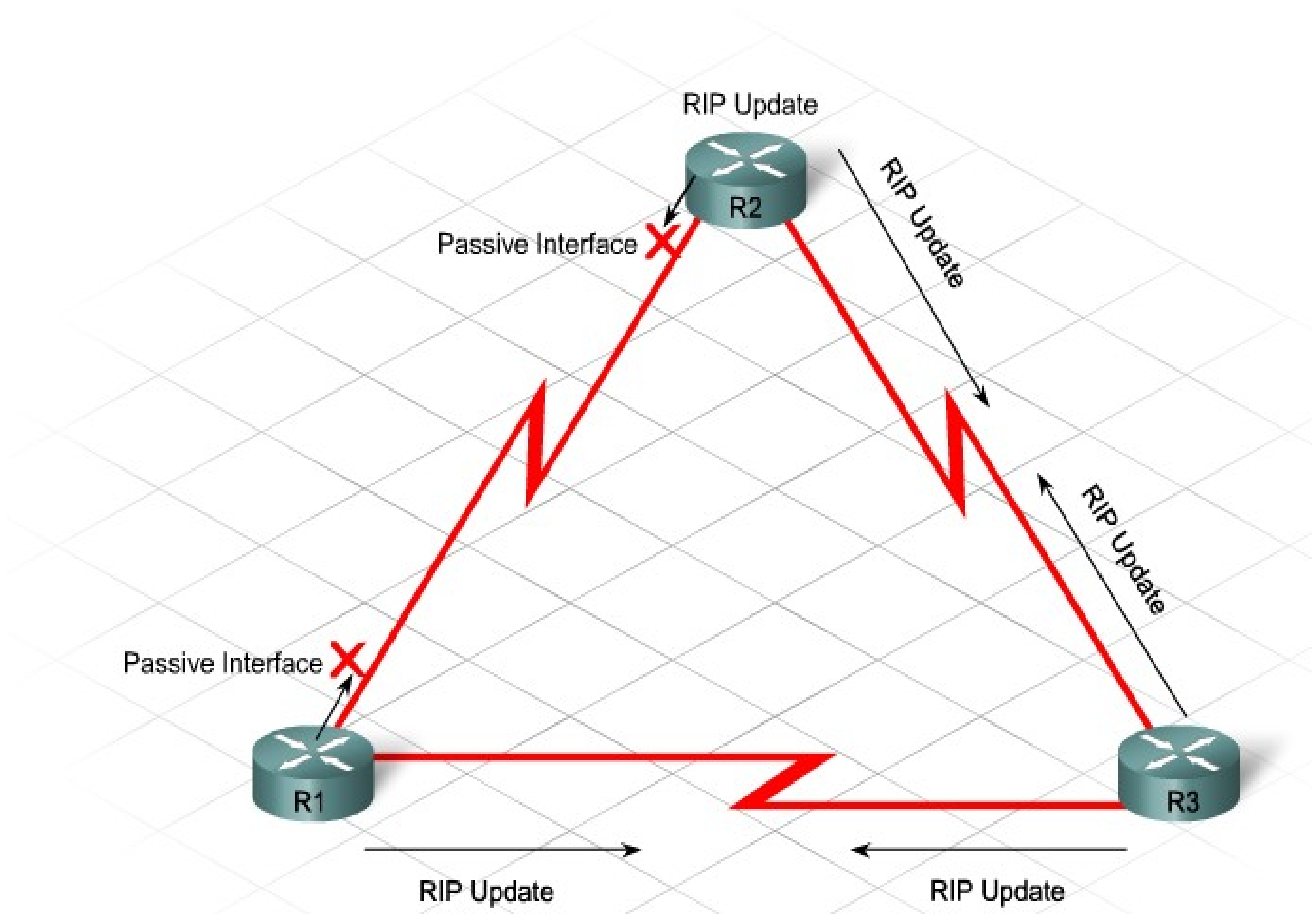
- To configure MD5 authentication, go to any interface participating in RIPv2 and enter the **ip rip authentication mode md5** command.
- This command encrypts any updates sent out that interface.



Problems on RIP

- Various performance and security issues arise when using RIP.
- The first issue concerns routing table accuracy.
- Both versions of RIP automatically summarize subnets on the classful boundary.
- Enterprise networks typically use classless IP addressing and a variety of subnets, some of which are not directly connected to each other, which creates discontiguous subnets.
- Unlike RIPv1, with RIPv2 the automatic summarization feature can be disabled. Using no auto-summary command.
- **Router(config-router)#no auto-summary**

- Another issue to consider is the broadcast nature of RIP updates.
- RIP immediately begins to send advertisements out all interfaces that belong to that network.
- These updates may not be needed on all portions of a network.
    - It produce unnecessary traffic.
    - Routing update intercepted, this makes the network less secure.
- The passive-interface command, issued in interface mode, disables routing updates on specified interfaces.
- **Router(config-router)#passive-interface interface-type interface-number**
- When the number of interfaces advertising RIP routes is **limited**, security and traffic control increase.


RIP Update
R2
Passive Interface
RIP Update
RIP Update
Passive Interface
R1
R3
RIP Update
RIP Update

Verifying RIP

- There are many show commands to assist the technician in verifying a RIP configuration and troubleshooting RIP functionality.
- The **show ip protocols** and **show ip route** commands are important for verification and troubleshooting on any routing protocol.
- The following commands specifically verify and troubleshoot RIP:

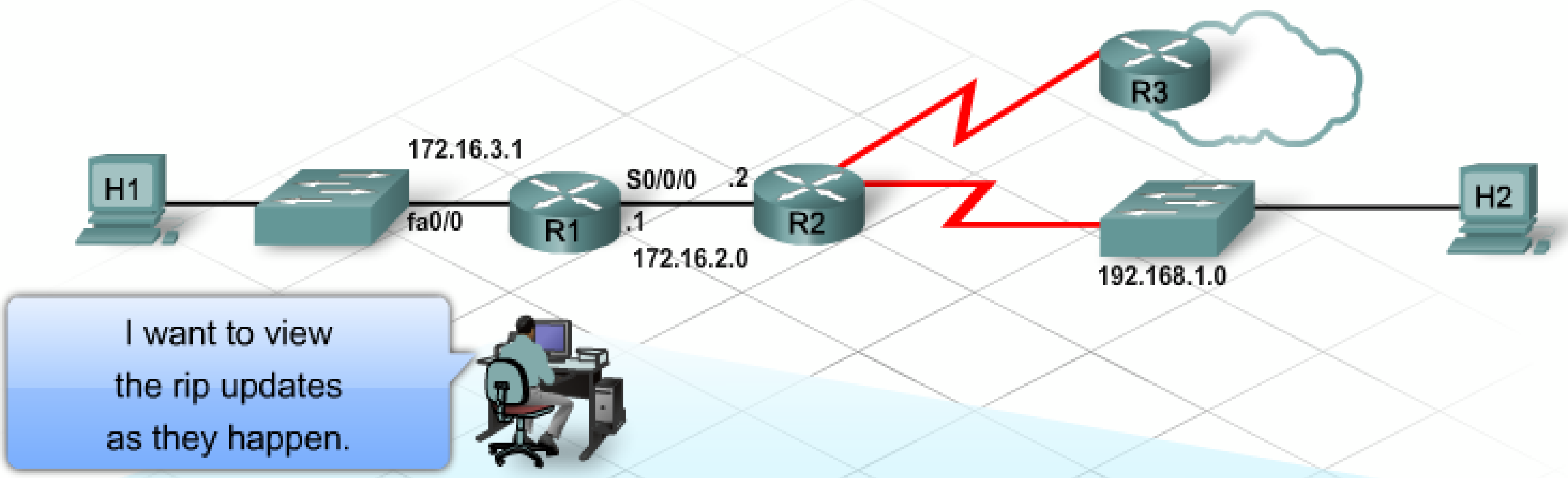
○ **show ip rip database**:

- Lists all the routes known by RIP

○ **debug ip rip**

○ **Undebug all**

- Displays RIP routing updates as sent and received in real time




```
R1#debug ip rip
RIP protocol debugging is on
R1#
*Aug 29 12:04:41.653: RIP: sending v1 update to 255.255.255.255 via FastEthernet0/0 (1
*Aug 29 12:04:41.653: RIP: build update entries
*Aug 29 12:04:41.653: subnet 172.16.1.0 metric 2
*Aug 29 12:04:41.653: subnet 172.16.2.0 metric 1
*Aug 29 12:04:41.653: network 192.168.1.0 metric 1
*Aug 29 12:04:43.188: RIP: sending v1 update to 255.255.255.255 via Serial0/0/0 (172.1
*Aug 29 12:04:43.188: RIP: build update entries
```

RIPv1 & RIPv2

	RIPv1	RIPv2	Both
1. Automatic route summarization			✓
2. No authentication	✓		
3. Hop-count metric			✓
4. Default 30-second update interval			✓
5. Administrative distance of 120			✓
6. Supports VLSM		✓	
7. Sends subnet mask in routing updates		✓	
8. Uses route poisoning, poison reverse, split horizon and holddowns to avoid loops			✓
9. Broadcast updates	✓		

DHCP (Dynamic Host Configuration Protocol)

- Dynamic Host Configuration Protocol (DHCP) is a network protocol that provides **automatic IP addressing** and other information to clients:

  IP address

  Subnet mask (IPv4) or prefix length (IPv6)

  Default gateway address

  DNS server address

- Available for both IPv4and IPv6

DHCP

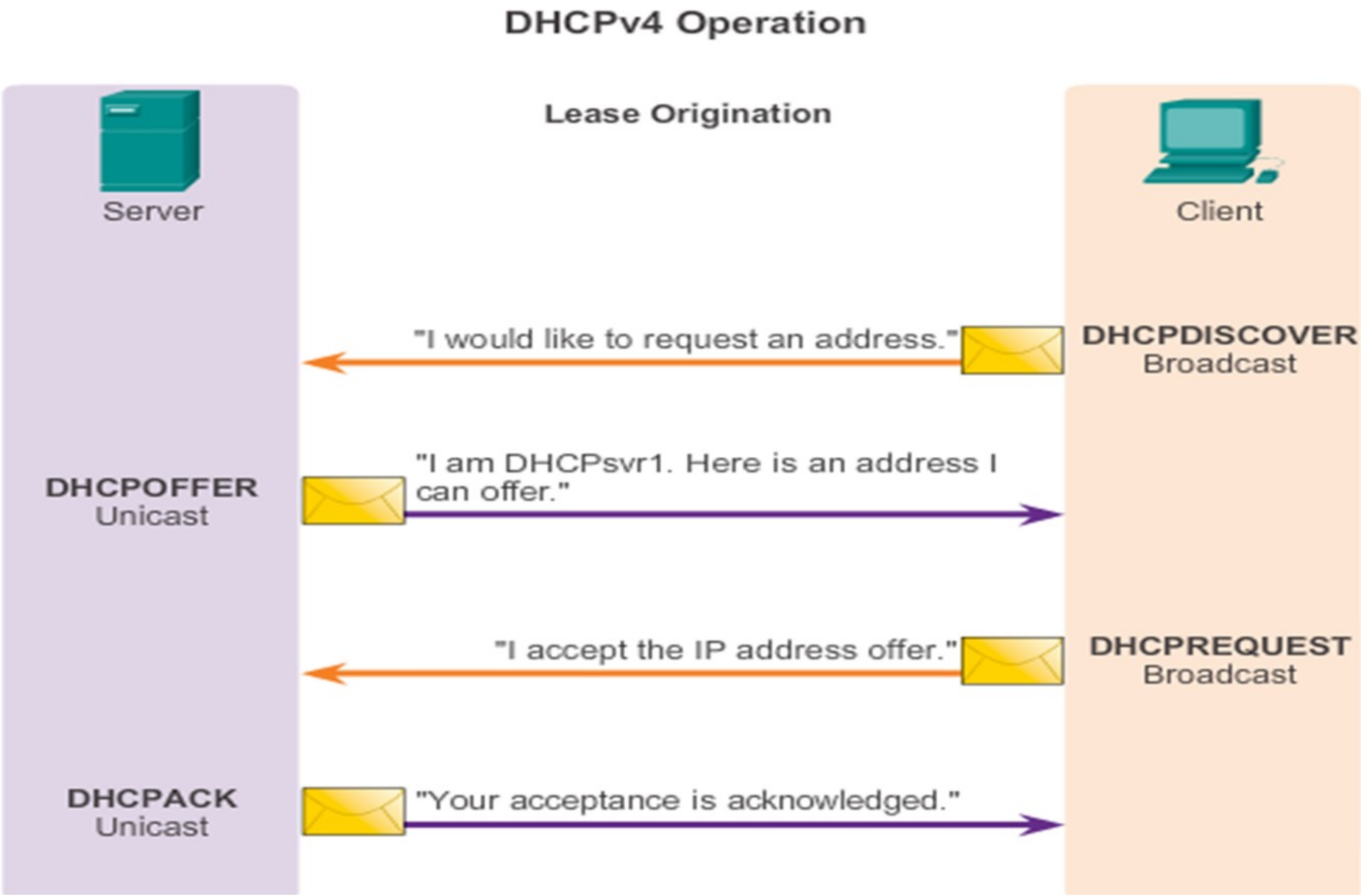
- DHCP uses three different address allocation methods

Manual Allocation - The administrator assigns a pre-allocated address to the client, and DHCP communicates only the IPv4 address to the device.

Automatic Allocation - DHCP automatically assigns a static IPv4 address permanently to a device, selecting it from a pool of available addresses. No lease.

Dynamic Allocation - DHCP dynamically assigns, or leases, an IPv4 address from a pool of addresses for a limited period of time chosen by the server, or until the client no longer needs the address. Most commonly used.

DHCP



DHCPv4 Operation
Lease Origination
Server
Client
"I would like to request an address."
DHCPDISCOVER
Broadcast
DHCPOFFER
Unicast
"I am DHCPsvr1. Here is an address I can offer."
"I accept the IP address offer."
DHCPREQUEST
Broadcast
DHCPACK
Unicast
"Your acceptance is acknowledged."

- Using a router configured with DHCP simplifies the management of IP addresses on a network; the administrator only needs to update a single, central router when IP configuration parameters change.

❑ There are eight basic steps to configuring DHCP via CLI

- 1. Create DHCP Address Pool
- 2. Specify the Subnet
- 3. Exclude IP Addresses
- 4. Specify the Domain Name
- 5. DNS Server IP Address
- 6. Set the Default Router
- 7. Set the Lease Duration
- 8. Verify the Configuration

1. Create DHCP Address Pool

```
Router(config)# ip dhcp pool LAN-address
Router(dhcp-config)#
```

Step 1: Create DHCP Address Pool

Navigate to the privileged EXEC mode, enter the password if prompted and then enter the global configuration mode. Now create a name for the DHCP server address pool. More than one address pool can exist on a router. The Cisco IOS CLI will enter the DHCP pool configuration mode. Use these commands:

```
Router> enable
Router# configure terminal
Router(config)# ip dhcp pool LAN-address
```

This example created an address pool named "LAN-address"

2. Specify the network

Router (dhcp-config)# network 172.16.0.0 255.255.0.0

Step 2: Specify the Network or Subnet

Specify the network or subnet network number and the subnet mask of the DHCP address pool. Use this command:

```
Router(dhcp-config)# network 172.16.0.0 255.255.0.0
```

Depending on the version of IOS, the subnet mask may also be specified using the prefix convention /16.

3. Excluding IP Address

Router (config)# ip dhcp excluded-address 172.16.1.100 172.16.1.103

Step 3: Exclude IP Addresses

Recall that the DHCP server assumes that all other IP addresses in a DHCP address pool subnet are available for assigning to DHCP clients. Exclude addresses from the pool so the DHCP server does not allocate those IP addresses. If a range of addresses is to be excluded, only the starting address and ending address need to be entered. Use this command:

```
Router(config)# ip dhcp excluded-address 172.16.1.100 172.16.1.103
```

The example shown excludes the four addresses, 172.16.1.100, 172.16.1.101, 172.16.1.102, and 172.16.1.103 from being given out to hosts by DHCP. These addresses can be statically assigned by the administrator.

4. Specify the domain name

```
Router (dhcp-config)# domain-name cisco.com
```

Step 4: Specify the Domain Name

Now specify the domain name for the client. Use this command:

```
Router(dhcp-config)# domain-name cisco.com
```

Clients in this example will receive the domain name cisco.com as part of their DHCP configuration. Domain name is an optional DHCP configuration parameter and is not necessary for DHCP to function. The network administrator can provide information as to whether or not a domain name is necessary.

DNS server Ip address

```
Router (dhcp-config)# dns server 172.16.1.103 172.16.2.103
```

Step 5: DNS Server IP Address

Now specify the IP address of a DNS server that is available to a DHCP client. One IP address is required. Up to eight IP addresses can be configured on one line. If listing more than one DNS Server list the servers in order of importance. Use this command:

```
Router(dhcp-config)# dns server 172.16.1.103 172.16.2.103
```

In this example, there are two DHCP servers that clients can use, a primary server and a secondary server. At least one DNS server must be configured for hosts to resolve host names and URLs in order to access services on the network.

6. Set the default gateway

```
Router (dhcp-config)# default-router 172.16.1.100
```

Step 6: Set the Default Gateway

Now specify the IP address of the default router for the DHCP clients on the network. Typically this will be the LAN IP of the router. This command will set the default gateway for the client devices on the network that will be using DHCP. After a DHCP client has booted, the client begins sending packets to its default router. The IP address must be on the same subnet as the client IP addresses given out by the router. One IP address is required. Use this command:

```
Router(dhcp-config)# default-router 172.16.1.100
```

Clients in this example use the router interface 172.16.1.100 as their default gateway.

7. Set the lease duration

```
lease {days [hours] [minutes]| infinite}
end
```

Step 7: Set the Lease Duration

DHCP gives out IP address information each time a host powers on and connects to the network. The default time that a client IP address is reserved for a specific host is one day. If the host does not renew its address, then the reservation ends and the IP address is again available to be given out through DHCP. It is possible to change the lease timer to a longer period of time, if necessary. This is the last step in configuring a DHCP service on a router. Use the end command to finish the DHCP configuration and return to the Global configuration mode. Use these commands:

```
lease {days [hours] [minutes]| infinite}
end
```

8. Verify the configuration

Router# show running-config

Step 8: Verify the Configuration

Verify the DHCP configuration by viewing the running-configuration. To do this use the command:

```
show running-config
```

Here is an example of the DHCP part of the configuration running on a DHCP enabled router:

```
!
ip dhcp pool LAN-addresses
domain-name cisco.com
network 172.16.0.0 255.255.0.0
ip dhcp excluded-address 172.16.1.100 172.16.1.103
dns-server 172.16.1.103 172.16.2.103
default-router 172.16.1.100
lease infinite
!
```